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Key Points:

- The response of ionospheric electron density due to chorus waves is most pronounced in D-region and exhibits multiple time-scale variations
- Long-term evolution of ionospheric density due to repetitive chorus waves can be reproduced through time-averaged precipitation flux
- Green light and blue light emissions show different responses to repetitive chorus waves

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Chorus Wave-Induced Ionospheric Variability

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Abstract Electron microbursts are short-lived but intense electron precipitation and constitute a significant source of electron losses into the ionosphere and lower atmosphere. They are primarily produced by wave-particle interaction with chorus waves in the magnetosphere. We investigate the effect of chorus wave-induced electron precipitation on the ionospheric density variability and auroral emission production. The effect of individual microbursts on ionospheric changes is investigated by coupling the Wave-Particle Interaction model, Geant4 Energetic Particle Precipitation model/Auroral Transport Model, and ionospheric chemistry model. It is found that the response in the ionospheric density is highly dependent on altitude, due to various time scales of different processes involved, including chemistry recovery timescale, individual microburst time duration, and repetition period of chorus waves. Different temporal behaviors are found for green and blue line radiation due to competition of chorus repetition period and radiative timescales.

Plain Language Summary Short bursts of energetic electrons, called microbursts, are an important way that Earth's near-space environment loses particles to the atmosphere. These bursts are thought to be driven mainly by chorus waves in the magnetosphere, but their effects on the ionosphere and auroral emissions have not been fully understood. Using a set of coupled models, this study shows that microburst-driven precipitation can produce complex and altitude-dependent changes in ionospheric density. The results also show that green and blue auroral emissions respond differently because the timing of the incoming electron bursts competes with the natural timescales of atmospheric chemistry and light production. These findings help clarify how rapid space weather processes can shape the upper atmosphere and aurora phenomena.

1. Introduction

Whistler-mode chorus waves are electromagnetic emissions often observed in the Earth's magnetosphere (He et al., 2023; Li et al., 2009; Santolík et al., 2014), over the frequency range from 0.05 to 0.8 f_{ce} (electron gyro-frequency). They typically appear as a series of rising-tone discrete elements in the frequency-time spectrogram, with separation time from a fraction of a second to a second (Shue et al., 2015). Chorus waves play critical roles in both radiation belt dynamics and magnetosphere-ionosphere coupling. Scattering by chorus waves has been identified as the dominant cause of diffuse auroral precipitation (Ni et al., 2016; Thorne et al., 2010), which provides the main source of energy for the high-latitude nightside upper atmosphere (Newell et al., 2009) and thus leads to enhanced ionization (e.g., Marshall & Cully, 2020) and variations in atmospheric chemistry (e.g., Randall et al., 2015). Chorus waves are capable of precipitating electrons over a wide energy range from ring current (1–10 s keV) to radiation belt (100 s keV and above) (Turner et al., 2023), which leads to energy deposition at altitude spanning the mesosphere to E-region ionosphere and is one of the major contributors to ionospheric conductance at auroral and subauroral latitudes.

Each discrete chorus element is capable of producing an impulsive precipitation of energetic electrons known as electron microbursts. Taken together, microburst precipitation events span electron energy from ~30 keV to >1 MeV and typically last a few tenths of a second to a second (Shumko et al., 2021). The process of generating electron microburst precipitation cannot be captured by quasi-linear scattering (Chen et al., 2020). Previous simulations with self-consistent chorus waves (Hikishima et al., 2010) have demonstrated a one-to-one correspondence between microbursts of precipitating electrons and chorus elements, which attributes the formation of an individual microburst to an interaction with one chorus element. Attempts have been made to examine detailed characteristics of those individual microbursts using low-earth orbiting satellites and their association with chorus waves (e.g., Breneman et al., 2017; Mozer et al., 2018). The majority of microbursts are shown to have sizes

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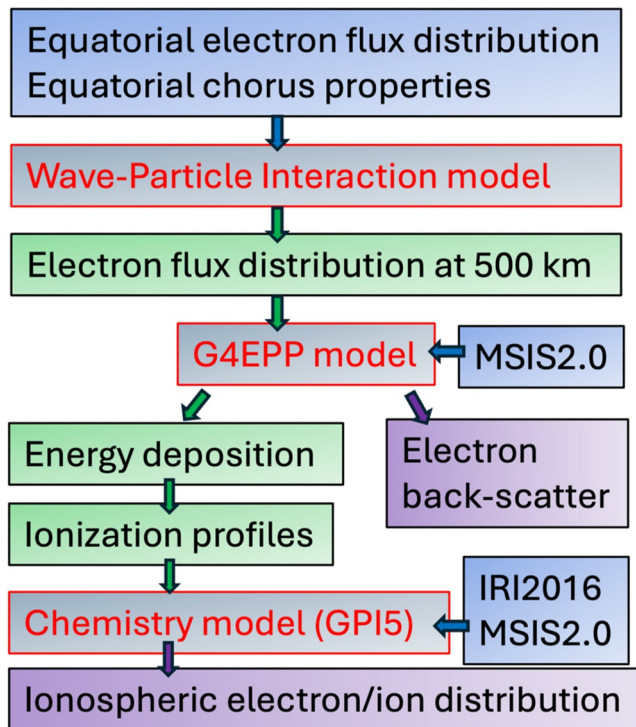


Figure 1. Illustration of the coupled WPI, G4EPP, and GPI5 models. Details are explained in the main text.

consistent with the sizes of chorus wave packets (Shumko et al., 2020) and have a characteristic repetition period consistent with that of chorus waves (Kandar et al., 2023).

The characteristic build-up timescale of electron precipitation on ionospheric density and ion composition ranges from a few seconds to tens of seconds, depending on altitude (Jones & Rees, 1973). The temporal morphology of auroral spectral emission features exhibits a nonlinear time response and depends on the time history of precipitation (Rees & Jones, 1973). Although the role of chorus waves in driving auroral electron precipitation (e.g., Thorne et al., 2010) and modulating global ionospheric conductance (e.g., Gillespie et al., 2024; Khazanov et al., 2025) is well recognized, previous studies have primarily employed time-averaged chorus wave properties. As a result, the ionospheric response to the highly bursty nature of chorus-driven electron precipitation remains largely unexplored, even though its characteristic timescales may be comparable to those governing ionospheric photochemistry. It remains unclear how the bursty electron precipitation affects the ionosphere and auroral emissions and how the chorus wave inherent repetition timescale affects the ionospheric responses. In this study, we aim at revealing the characteristics of ionospheric variabilities due to chorus-induced electron precipitation over various scales from sub-second, to seconds, to minutes, which covers some auroral emission radiative lifetime and chemistry process timescales.

2. Time-Dependent Ionospheric Response Model

Figure 1 illustrates the flowchart of coupled models for chorus-wave driven ionospheric responses, including the wave-particle interaction model (WPI) (Chen et al., 2020, 2021), Geant4 Energetic Particle Precipitation (G4EPP) model (Berland et al., 2023), and the GPI5 chemistry model (Kaeppeler et al., 2022) to investigate the effect of chorus waves in the magnetosphere on the ionosphere. The WPI model simulates the dynamics of electrons in a dipole magnetic field with prescribed chorus wave fields in the magnetosphere, and outputs electron unidirectional differential flux as a function of energy, pitch angle, and time at an altitude of 500 km, which will serve as the input for the G4EPP model. The G4EPP model is used to solve the energy deposit profile of a downward electron beam with a given energy and a given pitch angle into the ionosphere due to collision with neutrals, and then calculate the corresponding altitude profile of the electron-ion production rate, assuming 36 eV per electron-ion pair. The neutral mass density required for the G4EPP model is obtained from the MSIS2.0 (Mass Spectrometer Incoherent Scatter Radar-Empirical) model atmosphere Picone et al. (2002), which includes the atmospheric state (temperature, pressure, density) and constituent number densities. The G4EPP model (Berland et al., 2023) generates the ionization profiles for 285 electron beams over 19 energy values from 10 keV to 10 MeV and over 15 values of pitch angles from 0° to 70°. Using the responses of these reference electron beams, the ionization rate $q_e(z, t)$ driven by chorus wave-induced precipitation is generated through a linear superposition of these beams. The ionization rate is prescribed as a function of time and altitude z , and is then used to drive the GPI5 model, from which electron density variability as a function of time (t) and altitude (z) is solved.

The GPI5 model (named after its original four-species model developers Glukhov, Pasko and Inan (Glukhov et al., 1992)) is a simplified chemistry model in the D- and E-region with five species (Lehtinen & Inan, 2007): electron (n_e), light positive and negative ions (n^+ and n^-), and cluster positive and negative ions (n_x^+ and n_x^-), where n denotes the number density of a given species and the subscript x denotes the cluster species. The chemistry model involves the five coupled stiff first-order differential equations below (e.g., Kaeppeler et al., 2022)

$$\frac{dn_e}{dt} = S_0(z) + q_e(z, t) + \gamma n^- + \gamma_x n_x^- - \beta n_e - \alpha_d n_e n^+ - \alpha_d^c n_e n_x^+ \quad (1)$$

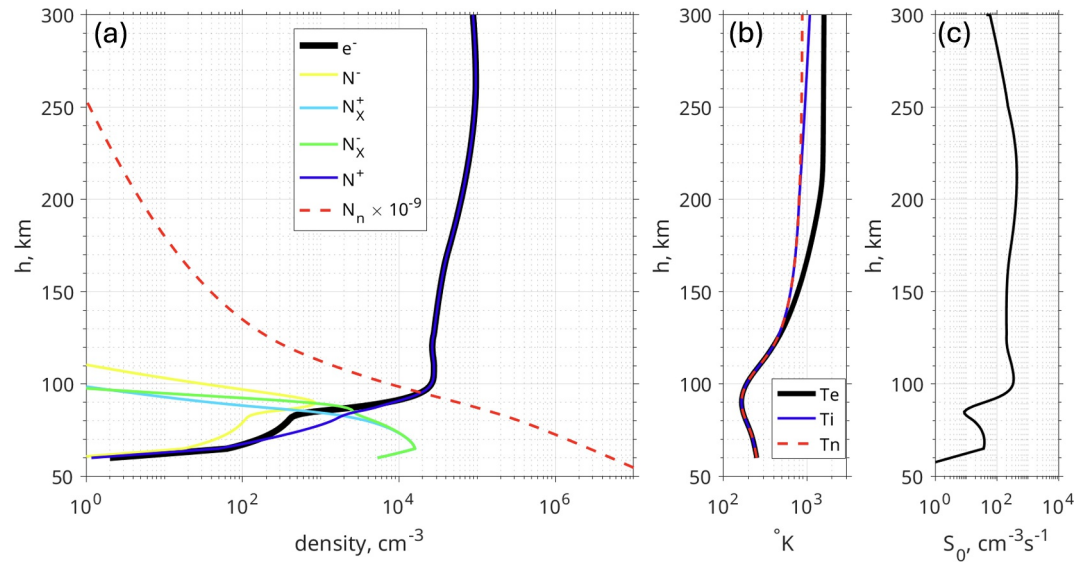


Figure 2. (a) Initial steady-state density profiles for the five charged species and the neutrals (n_n , scaled down by 10^{-9} for illustration purpose) altogether. (b) Initial temperature profiles for electrons (T_e), ions (T_i) and neutrals (T_n). (c) The altitude profile of the background ionization rate S_0 .

$$\frac{dn^-}{dt} = \beta n_e - \gamma n^- - \alpha_i n^- (n^+ + n_x^+) - \bar{X} n^- \quad (2)$$

$$\frac{dn_x^-}{dt} = -\gamma_x n_x^- - \alpha_i n_x^- (n^+ + n_x^+) + \bar{X} n^- \quad (3)$$

$$\frac{dn^+}{dt} = S_0(z) + q_c(z, t) - B n^+ - \alpha_d n_e n^+ - \alpha_i n^- n^+ - \alpha_i n_x^- n^+ \quad (4)$$

$$\frac{dn_x^+}{dt} = -\alpha_d^c n_e n_x^+ + B n^+ - \alpha_i n^- n_x^+ - \alpha_i n_x^- n_x^+ \quad (5)$$

The chemical coefficients (α , β , γ , B , and $\bar{X}$) are well-defined in Table 1 of Kaeppler et al. (2022) and their values are described in Section 2.2 of Kaeppler et al. (2022). $S_0(z)$ here is the time-independent background ionization rate while q_c is the ionization rate due to chorus-driven electron precipitation, which exhibits dependence on both time t and height z . The model describes the time evolution of ions and electrons at a given height z , with the assumption that the vertical transport is negligible, and thus the density variables at different altitudes are decoupled.

For the initial state ($t = 0$), the electron density profile n_e is from the 2016 International Reference Ionosphere (IRI2016) (Bilitza et al., 2017). A steady state is assumed at all heights initially (prior to the chorus wave effect), which is important for isolating the chorus-wave induced variation in the ionosphere. The steady state to maintain the initial electron density profile is established as follows. For a given z at $t = 0$, Equations 1–5 with $\frac{d}{dt} = 0$ and $q_c = 0$ are reduced to five nonlinear algebra equations, and are solved numerically through iterations to find the five unknown variables, the background ionization rate S_0 and the initial densities for the four species (n^- , n^+ , n_x^- and n_x^+). This procedure is repeated for a range of z from 1 km up to 500 km in the spacing of 1 km. For this study, we consider the ionosphere environment on 4:00 UT of 2018 May 8th (following Berland et al. [2023]'s study), at the location of geomagnetic latitude of 63.7° and geomagnetic longitude of 212°, corresponding to L -shell value of ~ 5.6 and magnetic local time of 3 hr (a region of intense chorus wave activities [e.g., He et al., 2023]).

Figure 2 shows the initial state of the ionosphere ($t = 0$), including the solution of the five unknowns (S_0 , n^- , n^+ , n_x^- and n_x^+). For the chemical equilibrium, light positive ion species dominate above 100 km, while other ion species become relevant for lower altitudes (Figure 2a). Electron temperature remains close to ion and neutral temperatures below 130 km while deviating from them at higher altitudes where electrons become essentially

collisionless (Figure 2b). The initial ionization rate S_0 shows a peak value of about $400 \text{ cm}^{-3} \text{ s}^{-1}$ around 100 km (Figure 2c), to maintain the electron profile prescribed by the IRI2016 model.

For examining photochemistry processes that lead to aurora emission as a result of chorus waves, we couple with the Auroral Transport Model (ATM) instead of G4EPP. The ATM model (Liang et al., 2026) is designed to simulate the transport of auroral electrons in the upper atmosphere, including impact ionization and secondary electron production, as well as the impact excitation of neutrals. The model computes the photon volume emission rates (VERs) of optical auroras at various wave-lengths in response to electron precipitation inputs and evolution of ionospheric density and composition. The excitation cross sections and the numerical scheme (two-stream approximation) to solve the electron transport equation are essentially inherited from the GLOW (GLObal airglOW) model (Solomon, 2017; Solomon et al., 1988), with a few notable advances: (a) A number of the reaction coefficients are updated and improved. (b) Changes in plasma density and temperature, as well as temperature-dependent reaction rates, are self-consistently calculated in the ATM, which allows a more realistic ionospheric response and photon emission profile due to chorus wave-induced electron precipitation. (c) Unlike the GLOW model which assumes the steady-state solution, the ATM model can be time-dependent, which is important for examining aurora produced by chorus waves with varying time-scales involved. The details of the model description can be found in Liang et al. (2016, 2024) and references within.

3. Simulation Results

Figure 3 shows the simulated electron flux at 500 km in the northern hemisphere in response to a single chorus element with frequency from 550 to 1,400 Hz over a duration of 0.2 s. The chorus wave propagates southward along the field line with an initial wave magnetic amplitude of 300 pT at $L = 5.5$ over a latitudinal range of -1° to -38° . The background dipole magnetic field is used with an equatorial cold electron density of 4.2 cm^{-3} , and the magnetospheric electron distribution used in Chen et al. (2020) is adopted. One can see in Figure 3a a leading microburst at $t = 0.4\text{--}1 \text{ s}$ with enhanced flux, mostly consistent with the energy-time region (enclosed by the white dashed line) expected by the gyro-resonant interaction with the prescribed chorus wave (Chen et al., 2020). The flux enhancement in the leading packet extends below the energy-time region as a result of electron backscattering. The backscattered electrons have an upward motion (with corresponding pitch angles above $\sim 120^\circ$), with flux 1–2 orders of magnitude less than the precipitation (Figure 3b). In the leading packet, both trapped and precipitation fluxes are enhanced, with a corresponding ionization profile of peak value $7,000 \text{ cm}^{-3} \text{ s}^{-1}$ near 100 km (Figure 3c). The subsequent packets are a manifestation of the bounce motion of trapped electrons, whose flux is also enhanced due to interaction with the chorus wave. The backscattered electrons within the leading packet in the northern hemisphere subsequently turn into precipitation in the conjugate (southern) hemisphere, where a new set of backscattered electrons are produced. The newly backscattered electrons arrive in the northern hemisphere as precipitation (pitch angle $< 60^\circ$) at the interval of 2–4 s, producing a secondary ionization profile (Figure 3c) with the ionization peak about three orders of magnitude smaller, and another set of backscattered electrons with negligible flux (pitch angle $> 120^\circ$ over the interval of 2–4 s in Figure 3b). In other words, the leading packet is a primary source of the ionization induced by the chorus wave, while the subsequent backscatter electron contribution is negligible.

Figure 4 shows electron density variation in the ionosphere due to chorus-induced ionization. Consider the first case with a single chorus element (Figures 4a–4c), a single burst of electron precipitation leads to impulsive ionization over the duration of the electron precipitation (a fraction of a second) over an altitude from 70 to 100 km (Figure 4a), with a peak ionization rate up to $700 \text{ cm}^{-3} \text{ s}^{-1}$. This chorus-driven ionization leads to an electron density enhancement at 80 km (Figure 4b) by a factor of 2, from 400 cm^{-3} to 900 cm^{-3} over a fraction of a second. When the burst ionization turns off ($q_e = 0$), a gradual recovery decay ensues from a state of elevated electron density toward the initial equilibrium state through the chemistry processes. At lower altitudes, the recovery timescale is as short as a few seconds, while it's about 10 s at 80–90 km and even longer at higher altitudes (Figure 4c). Figures 4d–4f demonstrates the second case with 10 repeated chorus elements over the first 5 s that are separated by 0.5 s. As shown, during the interval of chorus waves, the electron density experiences consecutive increases separated by the recovery decays during the gaps of chorus wave-induced precipitation, and thus appears as a sawtooth variation (Figure 4f, most pronounced over 70–100 km). Once the chorus activity recedes after 5 s, the electron density recovers to the initial equilibrium state over an altitude-dependent timescale as mentioned above. Figures 4g–4i shows the third case with repeated chorus elements that are separated by 0.5 s

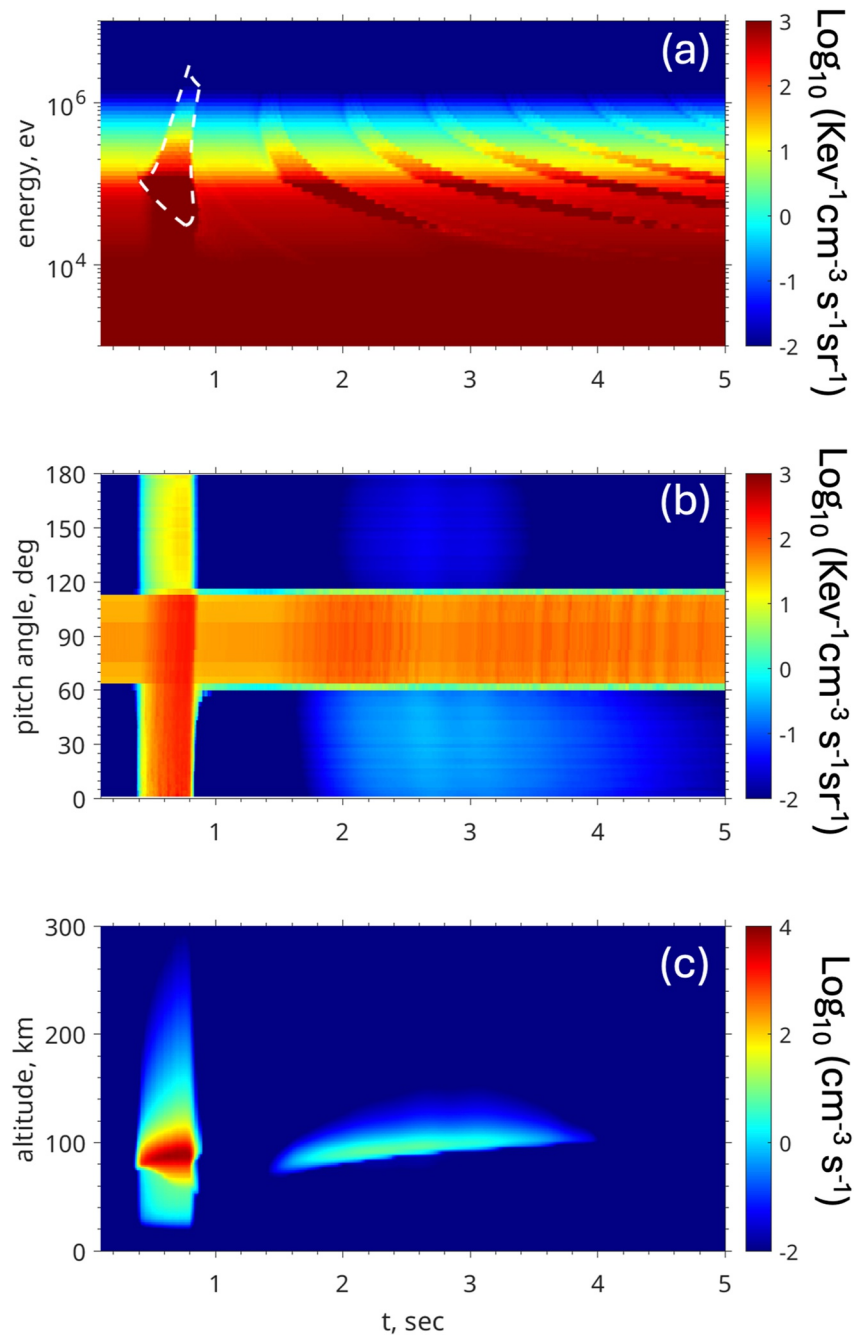


Figure 3. (a) WPI-G4EPP simulation of electron flux at a fixed altitude of 500 km in the northern hemisphere, due to interaction with a single southward-propagating chorus element, as a function of energy and time (b) as a function of pitch angle and time. The white dashed line in panel (a) delineates the theoretical energy-time area of the electron microburst due to resonance with the prescribed chorus wave. Panel (c) is the calculated ionization rate profile as a function of altitude and time.

over a sufficiently long interval (150 s). One can see electron density increases and then reaches a new steady state. The time required for reaching the new state, with a few seconds at 70 km and 10 s at 80–90 km, depends on the competition of the chemistry process and the ionization rate at that altitude. The new state will generate a sustained secondary peak at 80–90 km. The same steady-state solution can be reproduced using the time-averaged ionization rate $\langle q_c(z, t) \rangle$ (see the dashed lines in Figure 4i). In other words, for longer-term behavior than the chorus repetition timescale, one may use temporally averaged precipitation to obtain the long-term behavior. As

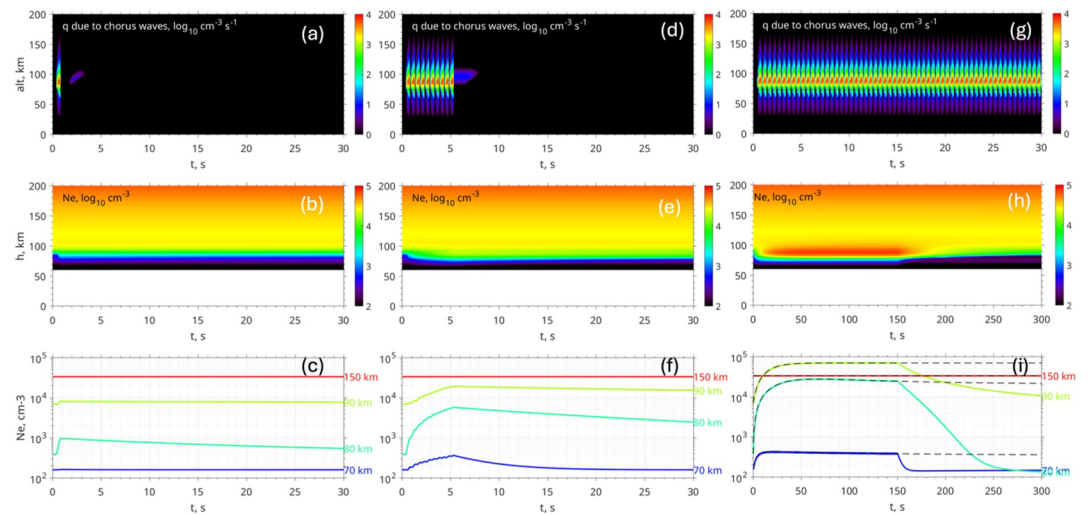


Figure 4. The left column shows the simulation results due to a single chorus element with (a) the chorus-induced ionization rate, (b) the simulated electron density profiles, and (c) the changes of electron density at selected altitudes (70, 80, 90, and 100 km). The middle column (d–f) shows the simulation results due to 10 consecutive chorus elements that are 0.5 s apart. The right column (g–i) shows the simulation results due to a train of chorus elements over the first 150 s that are 0.5 s apart. Panel (g) only shows repeated precipitation over the first 30 s, while panels (h–i) show the simulation results for 300 s. The dashed lines are the responses of electron density for constant time-averaged precipitation flux at corresponding altitudes, for comparison.

the chorus activity turns off at $t = 150$ s, the density recovers to its original steady state, with a recovery timescale range from a few seconds at 70 km to several minutes at 80 km, even longer at high altitudes. However, the variation of height-integrated total electron content variation, primarily due to D-region density variation, is only 0.1 TECU, which is about 5% of the initial value. This would make direct measurement of chorus-induced total electron content variation challenging. However, the signature of D-region density enhancement may be detected through radar measurement in a mode designed specifically for the D-region (e.g., Ma et al., 2022) or through ground-based observation of very low frequency radio wave amplitude from transmitters and receivers (e.g., Hendry et al., 2020).

It should be noted that as we only consider a short time response of ionospheric electron and ion densities, there is no vertical transport involved in the calculation, and no heating effect is included. One may expect that during the persistent electron precipitation due to extended chorus activity, effective electron heating in the E-region may occur where electrons and neutrals are less collisional. This effective heating may reduce the electron-ion recombination rate, leading to effective density enhancement. However, since the heat transfer between ionospheric thermal electrons and suprathermal electrons is most effective when the energy of the suprathermal electrons degrades to several eV (e.g., Schunk & Nagy, 1978), the heating rate is limited in the E-region for the precipitation energy range of interest in this study. At low altitudes in the D-region, the electron heating rate may become larger, but collisional cooling with dominant neutrals there suppresses electron temperature enhancement.

For examining the photon emission responses, we use the WPI mode to generate time-dependent electron precipitation flux, which is used to drive the ATM model (see Section 2). A photochemical equilibrium at the initial time is established prior to the effect of chorus waves. Figure 5 shows the photochemistry response due to the second case above with 10 chorus waves repeated every 0.5 s during the first 5 s. For the photochemistry equilibrium at $t = 0$ (prior to the effect of chorus waves), two zones of green light emission at wavelength 557.7 nm (with height-integrated column emission rate 250 R in Figure 5d) exist, a primary peak over 80–100 km (McDade et al., 1986) and a secondary peak over 130–160 km (Figure 5b). The blue line emission at wavelength 427.8 nm is rather weak initially with 10 R. On one hand, the blue light emission is produced via N_2^+ First Negative (1NG) band system with transitions from the upper electronic state $B^2\Sigma_u^+ \rightarrow X^2\Sigma_g^+$, with very fast radiative lifetime of ~ 60 ns. On the other hand, the 557.7 nm green light emission is a product of the relaxation process of O atom transition from the 1S state to the 1D state. The rise/decay timescale of the emission is

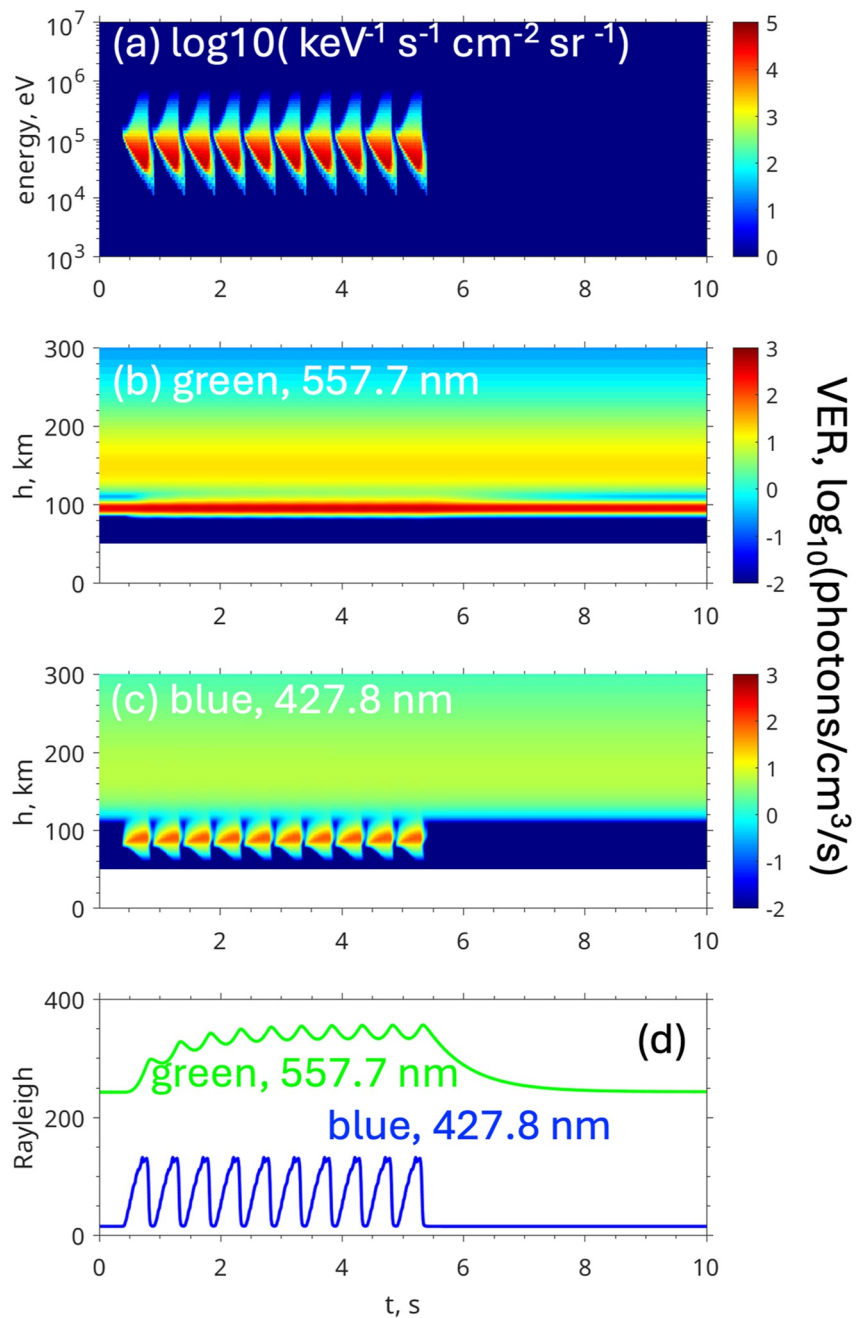


Figure 5. (a) Input electron flux (due to 10 chorus elements with 0.5 s spacing within the first 5 s) for the Auroral Transport Model simulation. Simulated Volume Emission Rate at (b) 557.7 nm (Green Line) and (c) 427.8 nm (Blue Line) as a function of height and time. Height-integrated column emission rates in unit of Rayleigh (10^6 photons/cm² s) are shown in panel (d).

determined by the Einstein probability of such a transition ($\sim 1.2 \text{ s}^{-1}$) and the quenching rate led by collisions with neutrals. In the altitude range of interest, The rise/decay timescale of the 557.7 nm emission is generally on order of a fraction of a second, comparable to the timescale of chorus wave separation and the timescale of chorus-driven electron precipitation. As chorus waves turn on, volume emission rates (VER) for both green and blue lines are enhanced, but over slightly different altitude ranges of 90–100 km and 80–100 km, respectively (Figures 5b and 5c). In addition, the two emissions respond differently with respect to time as shown in height-integrated emission rates (Figure 5d). The prompt blue line emissions follow closely in phase with the precipitation flux variation, with intensity variation of 100 R about the background 10 R. The blue line vanishes abruptly

in the absence of precipitation flux when chorus activity recedes. The green line radiation process, whose timescale is greater than the chorus repetition time, leads to a sawtooth increase of radiation intensity, accompanied by delayed phase with respect to the precipitation flux cycle. The green line emission gradually recovers to the initial level over the radiation timescale (0.8 s) when the chorus wave activity turns off.

It should be noted that the red line emission is negligibly small for this case of chorus-induced precipitation, due to the absence of <1 keV precipitation (Liang et al., 2016). The redline emission is produced from a cascading process from $O(^1S)$. $O(^1S)$ is a higher excitation state than $O(^1D)$. When $O(^1S)$ relaxes to $O(^1D)$, it emits 557.7 nm (green line), and when $O(^1D)$ further relaxes to ground state $O(^3P)$, it emits red line at 630 nm (with a radiative lifetime of 110 s).

4. Conclusions

We examine the effect of chorus-induced electron precipitation on ionospheric density changes and auroral emission variations from sub-second scale to minutes scale using the WPI-G4EPP-GPI5 model and the WPI-ATM model. Our principal conclusions are summarized as follows:

- The response of ionospheric electron density due to chorus waves is most pronounced in the D-region, with periodic sharp increases over sub-second scale, followed by chemistry recovery timescale from a few seconds to minutes.
- Long-term evolution of ionosphere electron density due to repetitive chorus waves can be reproduced through time-averaged electron precipitation flux.
- Green light and blue light emissions show different behaviors due to repetitive chorus waves. Blue light emissions closely follow the variation of chorus-induced electron precipitation flux, while green light emissions show accumulative increment above the background intensity.

The current study illustrates the fundamental processes governing the ionospheric response, including chemical and photochemical processes, to bursty chorus elements in the magnetosphere, where multiple characteristic time scales are involved. Although the simulations employ a specific configuration (e.g., particular ionospheric conditions and chorus wave properties), the general characteristics of the ionospheric response to repeated chorus waves are expected to remain valid.

The duration of electron precipitation (microbursts) is determined by chorus wave properties and their propagation in the magnetosphere (Chen et al., 2020). The repetition period of chorus waves is also expected to correspond to that of the associated electron microbursts. When the characteristic time scales of electron precipitation (i.e., the driver time scales) are much shorter than the time scale of ionospheric chemistry, the ionospheric electron density on chemical time scales or longer can be well approximated by the response to time-averaged electron precipitation, even though fine-scale density variations persist on the driver time scales. In contrast, when the driver time scales are much longer than the characteristic time scale of a given ionospheric process, such as the photochemistry responsible for the blue-line emission, the emission intensity closely follows the electron precipitation with little phase lag. When the driver and chemical time scales are comparable, as in the case of the photochemistry governing the green-line emission, the emission exhibits a delayed response relative to the precipitation driver.

It should be noted that the driver time scales can vary. Both the duration of electron microbursts and the repetition period of chorus waves tend to be larger on dayside than nightside (Shue et al., 2015; Shumko et al., 2021; Teng et al., 2017). Furthermore, the characteristic time scales of ionospheric chemistry depend strongly on altitude, and thus are affected by electron precipitation energy spectrum. In the present study, relatively hard electron precipitation is considered, resulting in energy deposition primarily in the D region. For softer electron precipitation, energy is deposited at higher altitudes, where chemical and photochemical time scales are generally longer. The energy spectrum of chorus-driven electron precipitation is controlled by the resonance condition for wave-particle interactions and therefore depends on both chorus-wave and background plasma properties. Softer electron precipitation may be favored when the cyclotron resonance shifts toward lower electron energies, for example, under conditions of higher chorus-wave frequencies or larger electron plasma-to-gyrofrequency ratios (e.g., Horne & Thorne, 2003).

Another interesting temporal scale is that of pulsating aurora, which typically exhibits on and off phases lasting a few seconds each (e.g., Nishimura et al., 2020). This characteristic period, on the order of 10 s, is comparable to

many ionospheric chemical and photochemical time scales and is therefore expected to produce strong coupling between magnetospheric forcing and ionospheric chemical responses.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The data in this study are provided in the data repository (Chen, 2026).

Acknowledgments

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